TRADEFLOW · PHASE 3B — EXIT LOGIC TEST

Is Window 5's Problem Really How Trades Are Managed, Not Which Ones Are Taken?

Testing the Score Floor Report's own diagnosis — same 6 fixed windows, floor 60 adopted as the 15m baseline

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The Score Floor Test found floor 60 captures essentially all available benefit from that lever, but explicitly could not move Window 5 (choppy, Jul-Dec 2023), which stayed near-flat across floor 50/60/65 (-16.3% → -14.9% → -14.8%). That report's conclusion was specific: Window 5's problem "is in how trades are managed once open, not which ones are taken" — an exit-logic problem, not an entry-quality one. This batch tests that diagnosis directly with two exit-logic conditions, isolated against the same 6 windows, floor 60 held constant.

Architecture Check — A Different Situation Than Score Floor

Unlike `SCORE_FLOOR`, which was a hardcoded constant requiring a parameterization fix before it could be tested at all, **`trailingStopMode` and `partialExit` were already proper `SimConfig` fields** with existing UI controls on `/simulate` (a Toggle and a set of Pills). No simulator.ts changes were needed to run this batch. The existing trade-management loop already implements: TP1 at 1R → move stop to breakeven → optionally close 50% (`partialExit`) → optionally trail the stop on whatever remains via ATR (`trailingStopMode: 'atr1'` or `'atr1.5'`) → TP2 (1.5-2.5R depending on conviction) and sometimes TP3 (3R) as further targets.

One honest approximation, flagged as instructed: Condition B asked for "trailing stop after 1R, target removed — exits only on trailing stop or reversal." That's not literally achievable with the existing architecture: **TP2 (and TP3, when assigned) remain hard exit triggers in the trade loop regardless of trailing mode** — there's no "no target" mode to switch into without deeper restructuring. The closest existing equivalent, used here as the minimal viable version: `partialExit: false` (no guaranteed profit lock at 1R) + `trailingStopMode: 'atr1'` (stop trails at 1x ATR once breakeven is reached). TP2/TP3 still exist as upside caps in this version — they're just rarely what closes the trade once Condition B's lower win rate is accounted for below.

Condition Comparison

CONDITION	DESCRIPTION	MEAN RETURN	MEAN PF	MEAN WR	PROFITABLE SUB-RUNS	MEAN MAX DD
A (baseline)	Floor 60, fixed exit (TP1 partial + fixed stop)	+20.4%	1.12	51.0%	10/18 (56%)	19.5%
B	Trailing after 1R, no partial lock	-16.3%	0.78	37.0%	3/18 (17%)	25.8%
C	Partial 50% at 1R + trail remainder	+18.2%	1.11	51.5%	11/18 (61%)	19.0%

> **Condition B is a clean, unambiguous failure** — every single metric got worse, not just on average but in every one of the 6 windows individually (see per-window table below). Win rate collapsed from 51.0% to 37.0%. The likely mechanism: on 15m's noisy candle structure, a 1x-ATR trail with no guaranteed partial lock gets whipsawed out shortly after the stop reaches breakeven, converting what would have been a banked 1R win into a breakeven-or-small-loss exit before the trade has room to develop. Removing the guaranteed partial lock at 1R was not a neutral change — it was load-bearing for this strategy on this timeframe.

Per-Window Return by Condition

WINDOW	REGIME	A (BASELINE)	B (TRAIL, NO PARTIAL)	C (PARTIAL+TRAIL)
1 (2020-10→2021-03)	Bull	+21.5%	-14.3%	+19.8%
2 (2023-10→2024-03)	Bull	+27.3%	-15.6%	+21.9%
3 (2021-05→2021-10)	Bear	+22.1%	-16.2%	+18.0%
4 (2022-04→2022-09)	Bear	+31.3%	-15.0%	+25.2%
5 (2023-07→2023-12)	**Choppy**	-14.9%	-30.0%	-12.0%
6 (2024-04→2024-09)	Choppy (sub.)	+35.2%	-6.4%	+36.0%

Window 5 Isolated — The Core Diagnostic

CONDITION	WINDOW 5 RETURN	WINDOW 5 PF	WINDOW 5 WR
A (baseline)	-14.9%	0.87	42.8%
B	-30.0%	0.58	33.3%
C	-12.0%	0.88	44.8%

> Condition C moves Window 5 from -14.9% to -12.0% — a real, positive 2.9-point improvement, with PF and win rate both also ticking up slightly. **It does not flip the window positive.** Condition B makes Window 5 dramatically worse, not better — the opposite of what the diagnosis predicted for this specific implementation.

Proportional Movement — Did Exit Logic Help Window 5 Disproportionately?

CONDITION	WINDOW 5 Δ (PP)	WINDOWS 1-4+6 AVG Δ (PP)	PATTERN
B	-15.0	-41.0	Hurt *other* windows far more than Window 5

C	+2.9	-3.3	Helped Window 5, slightly hurt the others

Condition C's pattern directionally confirms the original diagnosis, on a small scale. Window 5 moved positive (+2.9pp) while the other five windows moved slightly negative on average (-3.3pp) — a real divergence in the predicted direction: this specific exit change helped the choppy window relatively more than the trending ones, exactly what "Window 5's problem is exit management, not entry quality" would predict. **Condition B's pattern argues against its own version of the hypothesis** — removing the partial lock hurt the trending/bear windows (-35 to -46pp) far more violently than it hurt the already-weak Window 5 (-15pp), meaning whatever Condition B broke, it broke for a reason that has nothing to do with regime — it broke the core mechanism (the guaranteed 1R lock) that the whole strategy depends on, everywhere.

Trade-Off Check — Did Exit Logic Cap the Ceiling?

CONDITION	BEST INDIVIDUAL SUB-RUN	SYMBOL / WINDOW
A (baseline)	+108.0%	SOLUSDT, Window 6
B	+32.8%	SOLUSDT, Window 6
C	+108.0%	SOLUSDT, Window 6

Condition C does not cap the ceiling — its best individual result is identical to baseline (+108.0%, same symbol and window), meaning that specific run's price path either never triggered the ATR trail before reaching its full original target plan, or the trail and the original targets produced the same outcome. **Condition B caps the ceiling severely** (+32.8% vs +108.0% on the same symbol/window) — on top of failing broadly, it also gives up the upside that made the best cases so strong under the existing exit logic. There is no "better average, lower ceiling" trade-off to report for Condition C specifically — it doesn't show that pattern here; the cost is in the broader average (other windows -3.3pp), not the best case.

Average Hold Time

CONDITION	MEAN TRADE LENGTH (CANDLES)
A (baseline)	n/a — not captured when this data was originally recorded
B	7.89
C	7.91

Hold-time tracking was added for this batch; Condition A's saved data predates it and was not re-run to backfill, per the same discipline applied to the daily/weekly counter split in the prior report. B and C show essentially identical hold times (7.89 vs 7.91 candles) — the trailing mechanism isn't keeping trades open meaningfully longer in either configuration, which is itself informative: the failure mode in B isn't "trades ride too long and give it all back," it's something that happens quickly after breakeven.

Combined 15m + 1h Projected Impact

CONFIGURATION	COMBINED MEAN RETURN	WINDOWS ≥30%
Floor 60 + 1h (current best, fixed exit)	+23.6%	2/6
Condition C (partial+trail) + 1h	+22.5%	2/6

Condition C's combined projection is slightly worse than simply keeping the fixed exit (+22.5% vs +23.6%) — the small Window 5 gain (+2.9pp on 15m) is outweighed once blended with 1h by the slight degradation across the other five windows (-3.3pp average on 15m). Condition B is not worth projecting combined — its 15m collapse (-16.3% mean) would drag the blended number sharply negative regardless of 1h's strength.

Explicit Verdict

Condition B (trailing stop after 1R, no partial lock) failed decisively — every window got worse, win rate collapsed from 51.0% to 37.0%, and it broke the trending/bear windows far more violently than the choppy one, arguing against its own version of the diagnosis. **This condition is not recommended under any circumstance tested here.**

Condition C (partial 50% at 1R + trailing remainder) meaningfully improved Window 5 specifically, moving its return from -14.9% to -12.0%, while other windows moved slightly negative (-3.3pp average) — directionally confirming the diagnosis that Window 5's problem is exit management, not entry quality, but only on a small scale. It did not close Window 5 into positive territory, and combined with score floor 60 and the unchanged 1h baseline, blended mean return moves from +23.6% to +22.5% — a slight regression overall, not progress toward the 30% target.

Recommendation: do not adopt Condition C as tested — keep the fixed-exit floor-60 configuration as the working 15m baseline. The diagnosis from the Score Floor Test was directionally correct (Window 5's relative improvement under C versus the other windows proves the mechanism is real) but this specific exit-logic implementation (1x ATR trail) isn't strong enough to net-improve the strategy once its broader cost is counted. **This matches the brief's second scenario: exit logic helps Window 5 somewhat but it remains a stubborn weak point, and the net aggregate didn't improve — worth considering choppy-regime-conditional position sizing (reducing size specifically when the regime classifier detects ranging conditions) as the next, distinct lever, separate from both entry quality and exit mechanics tested so far.** One caveat worth carrying forward: only one trailing distance (1x ATR) was tested for Condition C; a looser distance (1.5x ATR) was not tried and might trade off differently — this finding is about the specific implementation tested, not a categorical proof that no exit-logic variant could ever help.

Appendix — All 54 Sub-Run Results (3 Conditions × 18 Runs)

WINDOW	SYMBOL	CONDITION	TRADES	WR	PF	RETURN	MAX DD	AVG HOLD (CANDLES)
1	BTCUSDT	A	164	52.4%	1.15	+18.5%	16.6%	n/a*
1	ETHUSDT	A	177	55.4%	1.07	+8.0%	12.2%	n/a*
1	SOLUSDT	A	137	56.2%	1.31	+38.0%	12.3%	n/a*
1	BTCUSDT	B	170	37.6%	0.72	-22.9%	28.9%	7.5
1	ETHUSDT	B	179	39.1%	0.69	-25.1%	25.3%	7.6
1	SOLUSDT	B	143	42.7%	1.05	+5.0%	14.6%	6.0
1	BTCUSDT	C	171	51.5%	1.05	+6.0%	20.2%	7.5
1	ETHUSDT	C	179	55.9%	1.11	+14.0%	10.9%	7.5
1	SOLUSDT	C	136	55.9%	1.34	+39.5%	12.4%	5.7
2	BTCUSDT	A	191	46.6%	0.77	-23.7%	24.6%	n/a*
2	ETHUSDT	A	171	49.1%	0.98	-2.0%	23.6%	n/a*
2	SOLUSDT	A	186	57.0%	1.68	+107.6%	15.6%	n/a*
2	BTCUSDT	B	199	34.7%	0.61	-31.6%	31.6%	9.2
2	ETHUSDT	B	189	38.1%	0.65	-26.2%	28.4%	8.0
2	SOLUSDT	B	198	39.9%	1.09	+10.9%	16.4%	7.0
2	BTCUSDT	C	190	46.8%	0.82	-21.8%	23.0%	9.1
2	ETHUSDT	C	181	50.8%	1.02	+2.9%	23.4%	8.0
2	SOLUSDT	C	186	57.0%	1.52	+84.6%	11.0%	7.2

3	BTCUSDT	A	181	47.0%	0.82	-20.7%	28.4%	n/a*
3	ETHUSDT	A	153	56.2%	1.34	+50.8%	24.6%	n/a*
3	SOLUSDT	A	149	61.7%	1.33	+36.2%	14.1%	n/a*
3	BTCUSDT	B	189	32.3%	0.60	-32.4%	32.5%	8.5
3	ETHUSDT	B	157	43.3%	0.97	-3.3%	28.4%	8.2
3	SOLUSDT	B	152	40.1%	0.86	-12.9%	23.3%	7.0
3	BTCUSDT	C	184	46.7%	0.81	-22.4%	30.7%	8.5
3	ETHUSDT	C	157	56.7%	1.28	+40.9%	26.9%	8.4
3	SOLUSDT	C	150	61.3%	1.32	+35.5%	15.0%	7.1
4	BTCUSDT	A	177	45.8%	0.95	-6.7%	23.6%	n/a*
4	ETHUSDT	A	186	54.8%	1.39	+55.6%	17.5%	n/a*
4	SOLUSDT	A	201	53.2%	1.23	+45.1%	14.9%	n/a*
4	BTCUSDT	B	185	32.4%	0.63	-31.9%	33.2%	8.0
4	ETHUSDT	B	186	40.9%	0.93	-7.3%	22.4%	7.7
4	SOLUSDT	B	200	33.5%	0.95	-5.8%	19.6%	6.4
4	BTCUSDT	C	181	44.8%	0.86	-17.1%	24.8%	8.2
4	ETHUSDT	C	193	54.4%	1.31	+43.2%	14.8%	7.7
4	SOLUSDT	C	189	55.0%	1.28	+49.4%	14.9%	6.6
5	BTCUSDT	A	172	43.0%	0.93	-9.1%	23.3%	n/a*
5	ETHUSDT	A	150	41.3%	0.79	-21.5%	24.7%	n/a*
5	SOLUSDT	A	188	44.1%	0.89	-14.2%	23.3%	n/a*
5	BTCUSDT	B	179	32.4%	0.63	-31.2%	32.5%	8.8
5	ETHUSDT	B	154	33.8%	0.47	-31.9%	34.0%	8.4
5	SOLUSDT	B	189	33.9%	0.64	-26.8%	27.3%	7.9
5	BTCUSDT	C	175	42.9%	0.92	-10.9%	20.9%	8.7
5	ETHUSDT	C	154	42.9%	0.74	-23.4%	27.3%	8.4
5	SOLUSDT	C	193	48.7%	0.99	-1.8%	18.2%	8.1
6	BTCUSDT	A	174	54.6%	1.21	+26.7%	12.9%	n/a*
6	ETHUSDT	A	170	41.8%	0.69	-29.0%	29.5%	n/a*
6	SOLUSDT	A	194	58.0%	1.66	+108.0%	10.0%	n/a*
6	BTCUSDT	B	177	37.9%	0.81	-18.8%	20.6%	8.2
6	ETHUSDT	B	181	31.5%	0.51	-33.2%	33.7%	8.6
6	SOLUSDT	B	204	41.4%	1.29	+32.8%	11.6%	8.9
6	BTCUSDT	C	180	53.3%	1.18	+22.8%	14.9%	8.1
6	ETHUSDT	C	181	44.8%	0.82	-22.7%	23.9%	8.6
6	SOLUSDT	C	200	56.8%	1.66	+108.0%	8.7%	9.2

* Condition A predates hold-time tracking, added for this batch.

Report generated from 36 newly-run live simulations (Conditions B and C, 18 each) plus the Score Floor Test's existing 18 floor-60 fixed-exit runs (Condition A), against real historical Binance market data. Zero console errors across both new conditions. Raw data: `training-`

`exports/phase3b-conditionB-results.json`, `training-exports/phase3b-conditionC-results.json`, `training-exports/phase3-floor60-results.json`.